Computations Performed at UniBwM

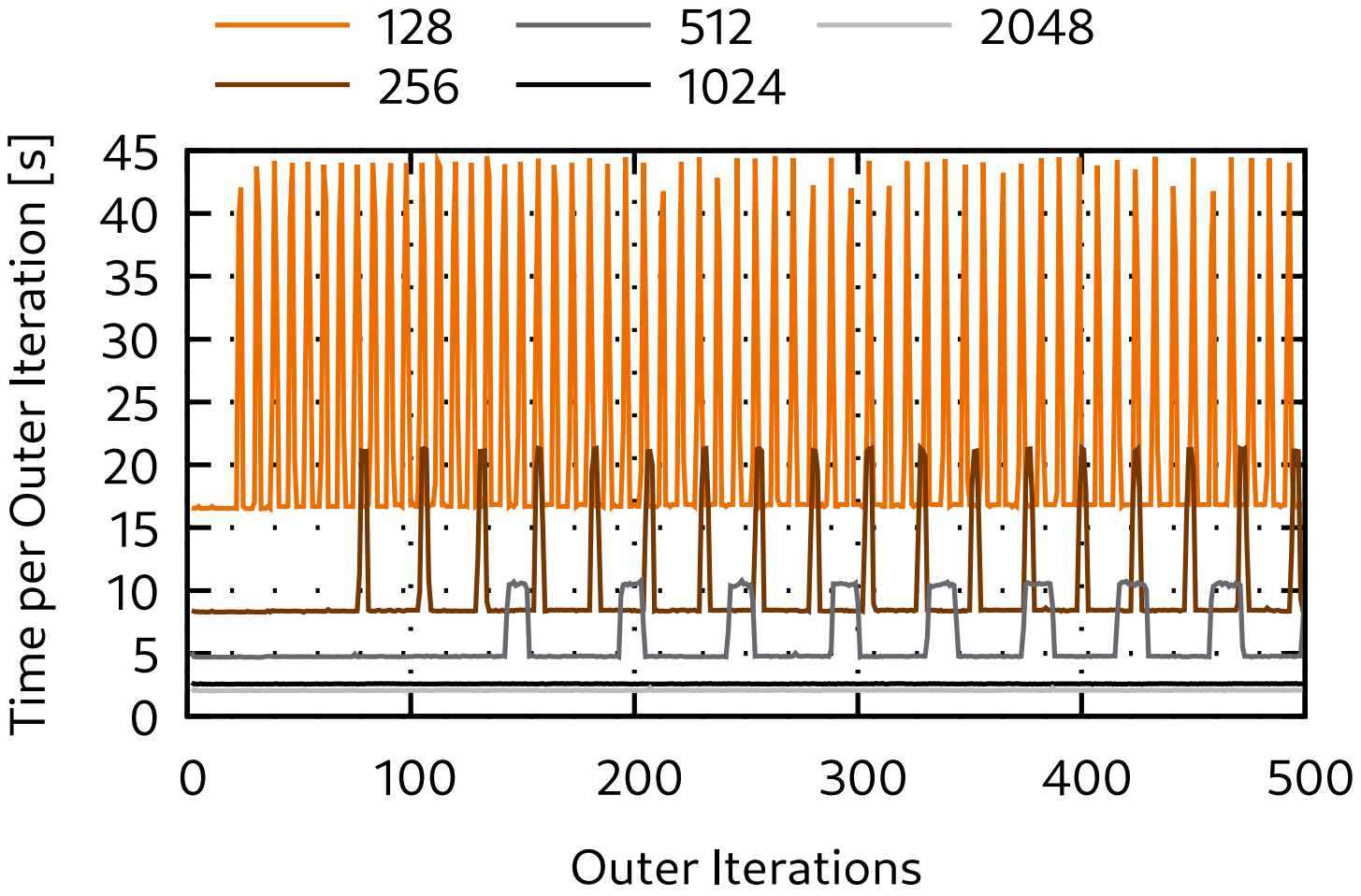
Eike Tangermann

Universität der Bundeswehr München
Germany

OHC-2 – Guimarães– 01. July 2026

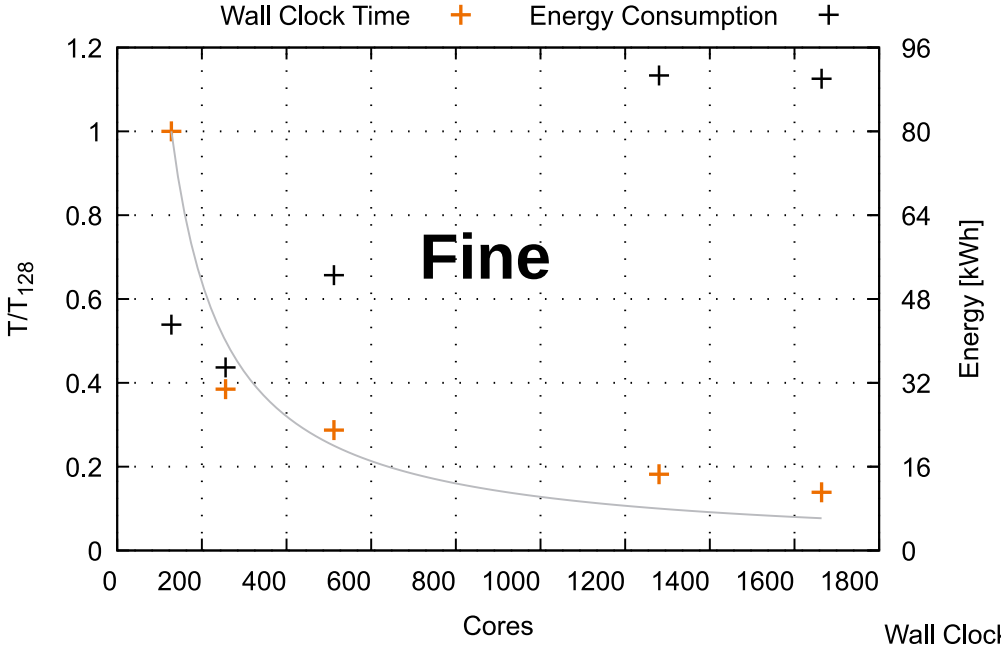
Last year: Time Variation

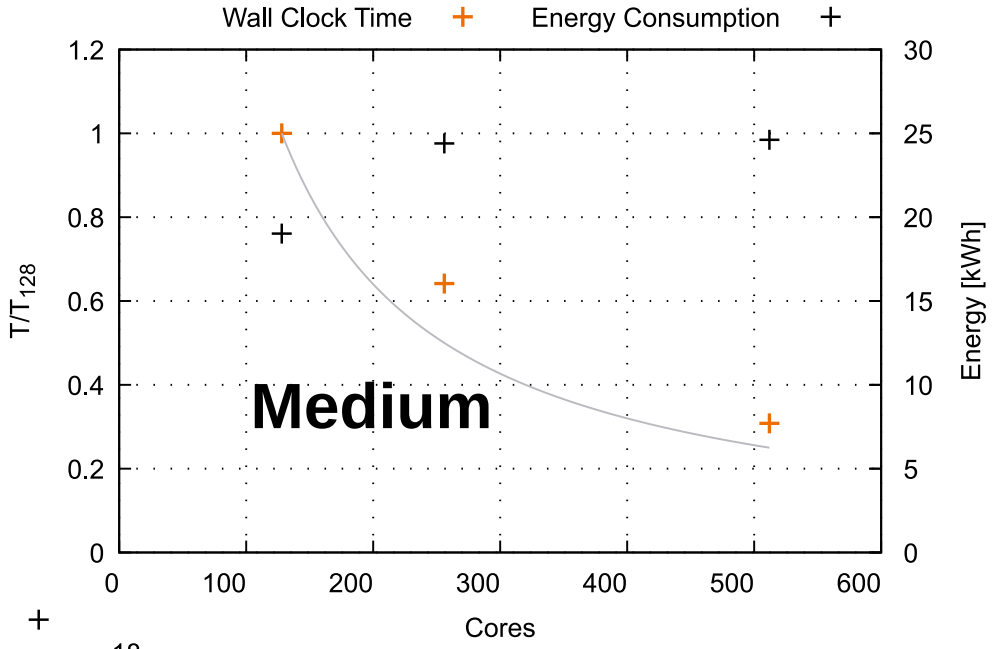
- Variations of iteration time
- Different on each node
- Still appears (in OHC cases)
- Not really a factor in other cases
- This year less nodes free for testing

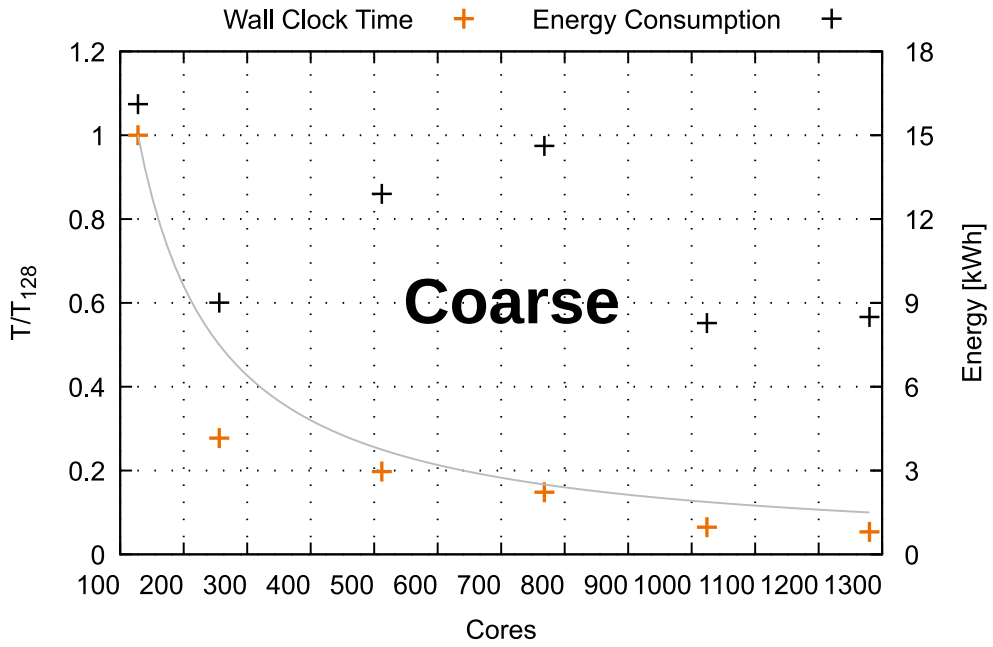


Scaling on different meshes

Universität der Bundeswehr München
Professur für Numerische Methoden in der Luft- und Raumfahrttechnik



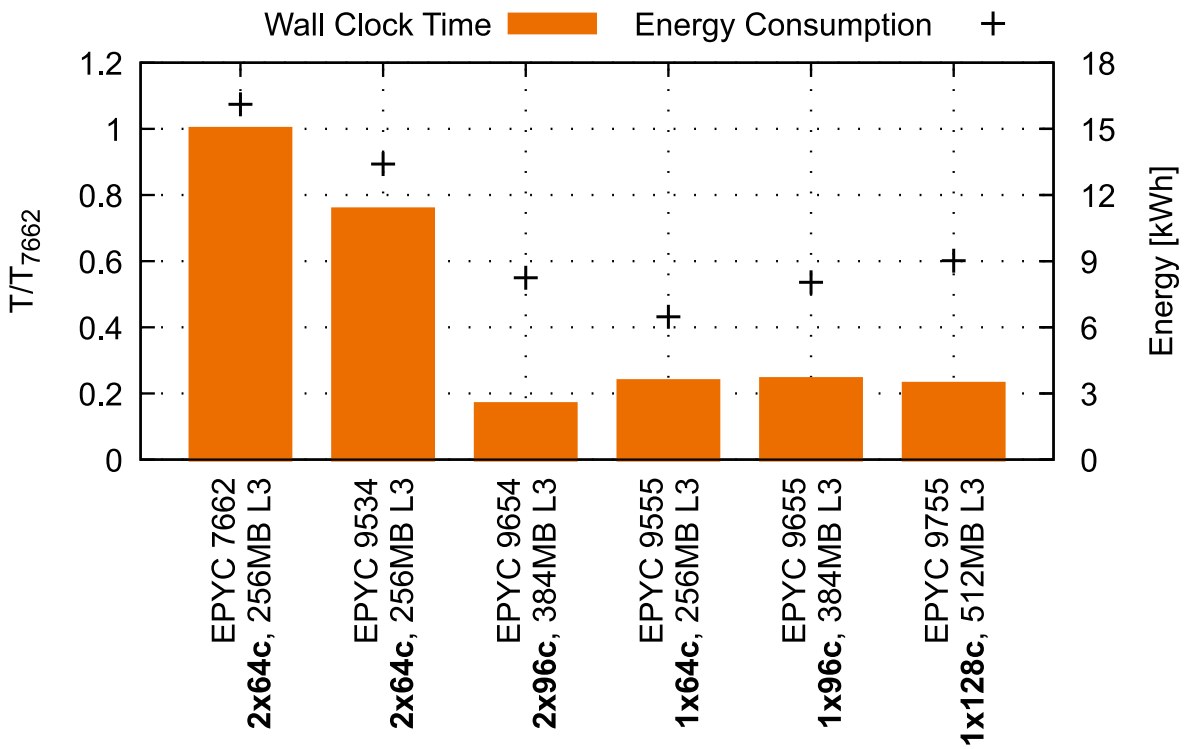




CPU Variations

- Comparison of different EPYC CPUs
- Zen 2 (Rome) and 4 (Genoa) on site
- Zen 5 (Turin) made available by

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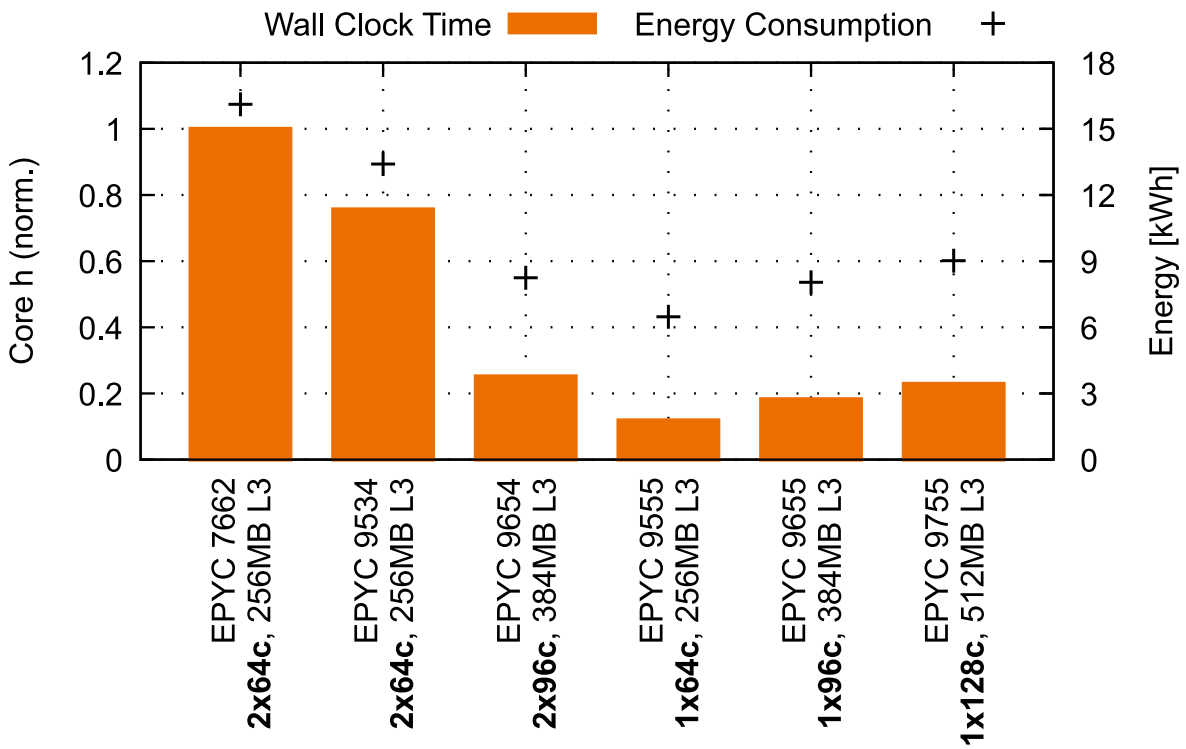


- Zen 4, 64c slightly faster than Gen 2
- Zen 4, 96 core much faster -> L3?
- All Zen 5 slightly faster than Gen 4 -> not L3
- Zen 5 not much difference in time from 64 to 128 cores
  -> compiler tuning necessary?

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Conclusions

- CPU variations show improvement with newer generations
- Suspected L3 effect not confirmed
- Gen 5 might benefit from specific compiler tuning
- Newer versions of EPYC do not show the time variations

